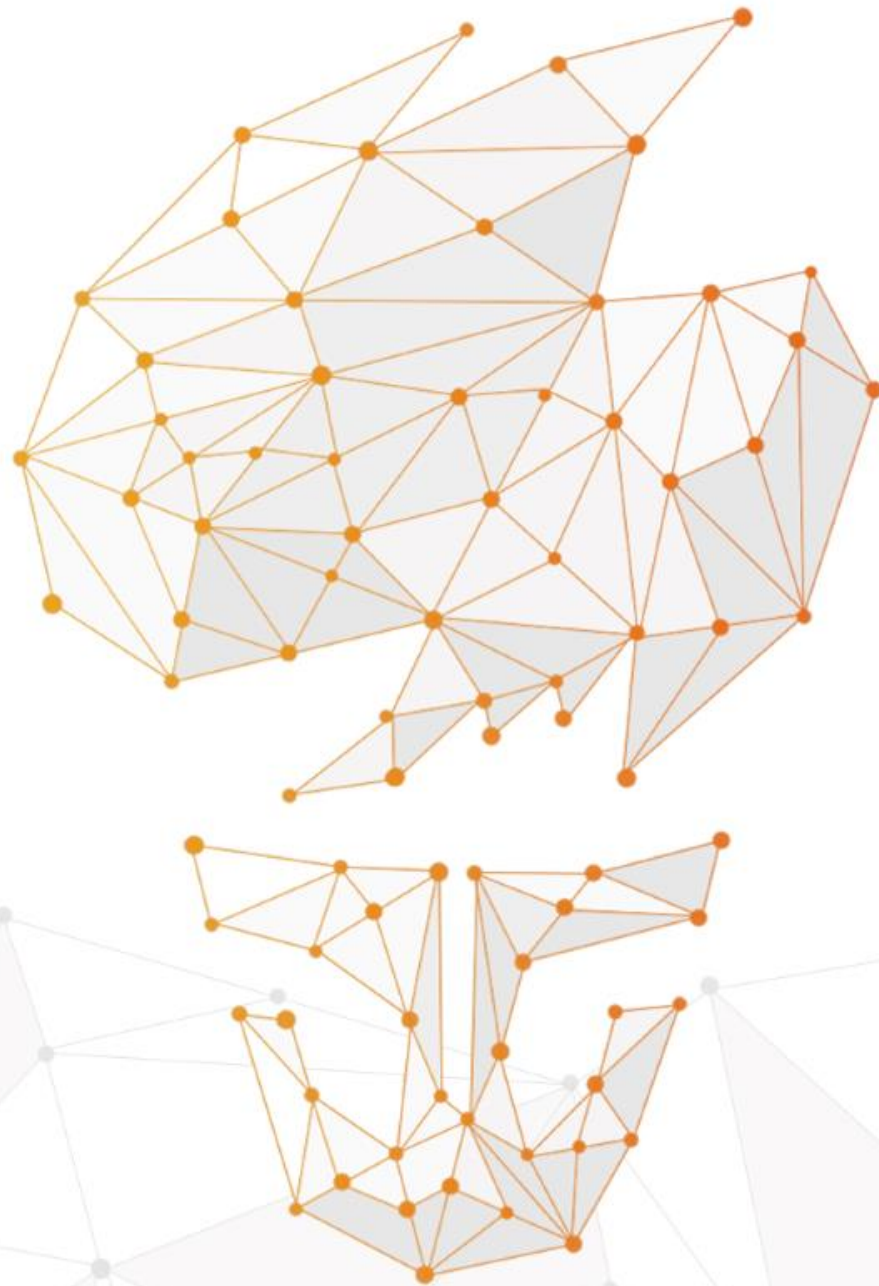


高数期中复习

致诚书院学长学姐带飞课

Chapter 10, Chapter 11

By 苏宇鹏 3.26 2023



Chapter12

向量与空间几何

DEFINITIONS The vector represented by the directed line segment $\overrightarrow{AB}$ has **initial point** A and **terminal point** B and its **length** is denoted by $|\overrightarrow{AB}|$. Two vectors are **equal** if they have the same length and direction.

DEFINITION

If $\mathbf{v}$ is a **two-dimensional** vector in the plane equal to the vector with initial point at the origin and terminal point (v_1, v_2) , then the **component form** of $\mathbf{v}$ is

$$\mathbf{v} = \langle v_1, v_2 \rangle.$$

If $\mathbf{v}$ is a **three-dimensional** vector equal to the vector with initial point at the origin and terminal point (v_1, v_2, v_3) , then the **component form** of $\mathbf{v}$ is

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle.$$

The **magnitude or length** of the vector $\mathbf{v} = \overrightarrow{PQ}$ is the nonnegative number

$$|\mathbf{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

(See Figure 12.10.)

向量运算

DEFINITIONS Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ be vectors with k a scalar.

Addition: $\mathbf{u} + \mathbf{v} = \langle u_1 + v_1, u_2 + v_2, u_3 + v_3 \rangle$

Scalar multiplication: $k\mathbf{u} = \langle ku_1, ku_2, ku_3 \rangle$

DEFINITION The **dot product** $\mathbf{u} \cdot \mathbf{v}$ (“ $\mathbf{u}$ dot $\mathbf{v}$ ”) of vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3.$$

向量投影: $\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}.$

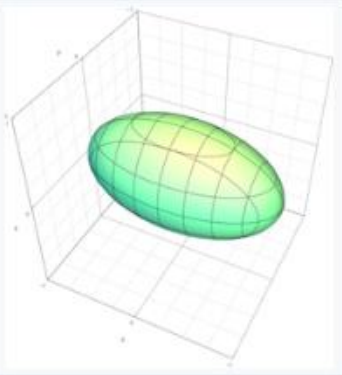
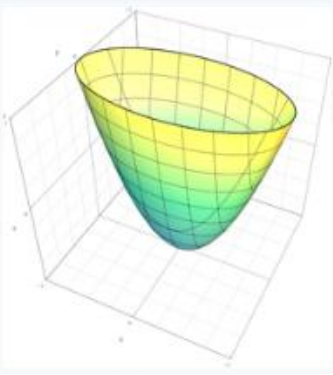
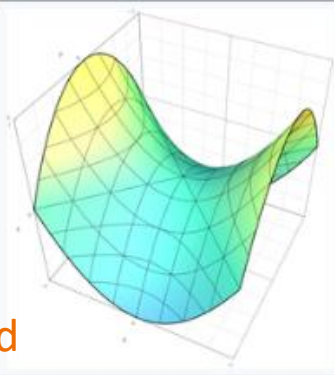
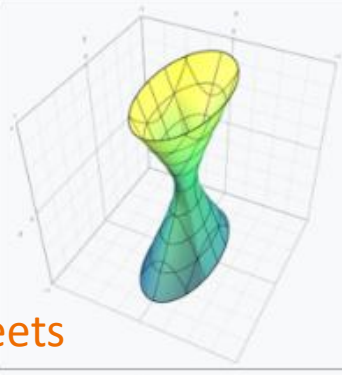
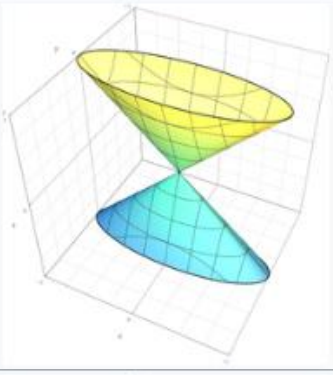
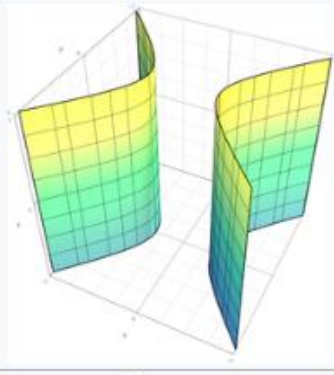
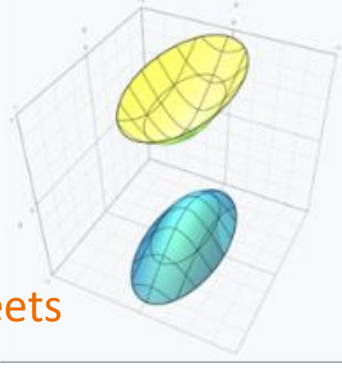
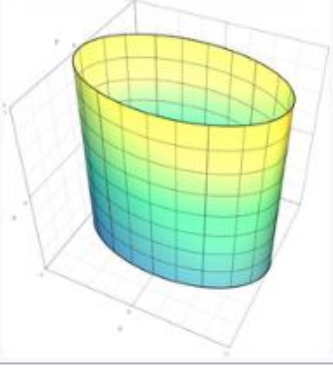
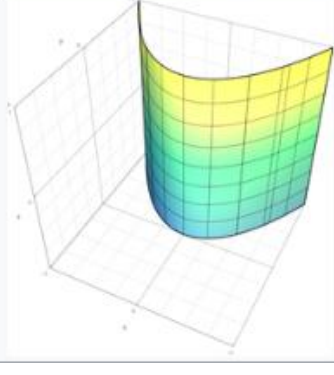
DEFINITION The **cross product** $\mathbf{u} \times \mathbf{v}$ (“ $\mathbf{u}$ cross $\mathbf{v}$ ”) is the vector

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}.$$
$$\mathbf{u} \times \mathbf{v} = (|\mathbf{u}||\mathbf{v}| \sin \theta) \mathbf{n}.$$

点到直线距离: $d = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|$

二次曲面

二次曲面 - 维基百科，自由的百科全书 (wikipedia.org)

$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ ellipsoid		$\frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0$ elliptical paraboloid		$\frac{x^2}{a^2} - \frac{y^2}{b^2} - z = 0$ hyperbolic paraboloid	
$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$ hyperboloid of one sheets		$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$ elliptical cone		$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ hyperbolic cylinder	
$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$ hyperboloid of two sheets		$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ elliptical cylinder		$x^2 + 2ay = 0$ parabolic cylinder	

Chapter13

向量函数与空间运动

向量函数

The vector $\mathbf{r}(t) = \overrightarrow{OP} = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$

from the origin to the particle's **position** $P(f(t), g(t), h(t))$ at time t is the particle's **position vector**

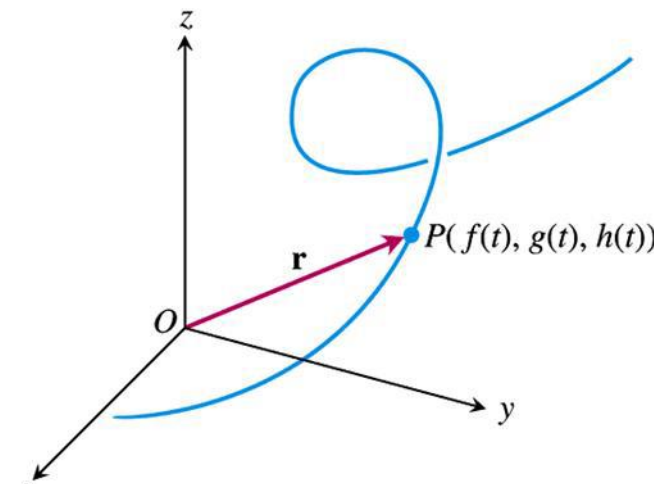
DEFINITION Let $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ be a vector function with domain D , and $\mathbf{L}$ a vector. We say that $\mathbf{r}$ has **limit** $\mathbf{L}$ as t approaches t_0 and write

$$\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{L}$$

if, for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all $t \in D$

$$|\mathbf{r}(t) - \mathbf{L}| < \epsilon \quad \text{whenever} \quad 0 < |t - t_0| < \delta.$$

DEFINITION A vector function $\mathbf{r}(t)$ is **continuous at a point** $t = t_0$ in its domain if $\lim_{t \rightarrow t_0} \mathbf{r}(t) = \mathbf{r}(t_0)$. The function is **continuous** if it is continuous at every point in its domain.



极限

连续性

向量函数的导数

DEFINITION The vector function $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ has a **derivative (is differentiable) at t** if f , g , and h have derivatives at t . The derivative is the vector function

$$\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{df}{dt}\mathbf{i} + \frac{dg}{dt}\mathbf{j} + \frac{dh}{dt}\mathbf{k}.$$

Differentiation Rules for Vector Functions

Let $\mathbf{u}$ and $\mathbf{v}$ be differentiable vector functions of t , $\mathbf{C}$ a constant vector, c any scalar, and f any differentiable scalar function.

1. *Constant Function Rule:* $\frac{d}{dt}\mathbf{C} = \mathbf{0}$

2. *Scalar Multiple Rules:* $\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$
 $\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$

3. *Sum Rule:* $\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$

4. *Difference Rule:* $\frac{d}{dt}[\mathbf{u}(t) - \mathbf{v}(t)] = \mathbf{u}'(t) - \mathbf{v}'(t)$

5. *Dot Product Rule:* $\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$

6. *Cross Product Rule:* $\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$

7. *Chain Rule:* $\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

向量函数的积分

DEFINITION The **indefinite integral** of $\mathbf{r}$ with respect to t is the set of all antiderivatives of $\mathbf{r}$, denoted by $\int \mathbf{r}(t) dt$. If $\mathbf{R}$ is any antiderivative of $\mathbf{r}$, then

$$\int \mathbf{r}(t) dt = \mathbf{R}(t) + \mathbf{C}.$$

DEFINITION If the components of $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$ are integrable over $[a, b]$, then so is $\mathbf{r}$, and the **definite integral** of $\mathbf{r}$ from a to b is

$$\int_a^b \mathbf{r}(t) dt = \left(\int_a^b f(t) dt \right) \mathbf{i} + \left(\int_a^b g(t) dt \right) \mathbf{j} + \left(\int_a^b h(t) dt \right) \mathbf{k}.$$

Ideal Projectile Motion Equation

抛体运动方程:

$$\mathbf{r} = (v_0 \cos \alpha)t \mathbf{i} + \left((v_0 \sin \alpha)t - \frac{1}{2}gt^2 \right) \mathbf{j}. \quad (5)$$

弧长与单位切向量

DEFINITION The **length** of a smooth curve $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$, $a \leq t \leq b$, that is traced exactly once as t increases from $t = a$ to $t = b$, is

弧长:
$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \int_a^b |\mathbf{v}| dt \quad (1)$$

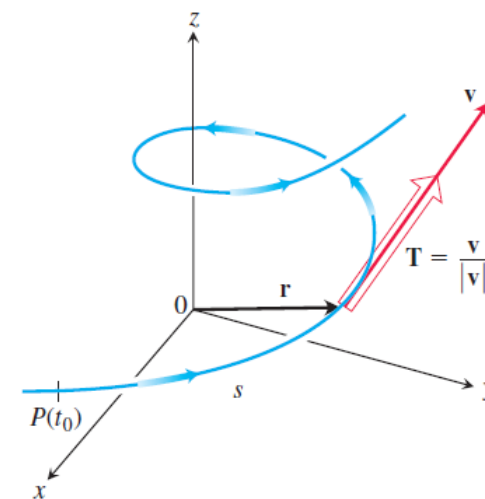
Arc Length Parameter with Base Point $P(t_0)$

$$s(t) = \int_{t_0}^t \sqrt{[x'(\tau)]^2 + [y'(\tau)]^2 + [z'(\tau)]^2} d\tau = \int_{t_0}^t |\mathbf{v}(\tau)| d\tau \quad (3)$$

We already know the velocity vector $\mathbf{v} = d\mathbf{r}/dt$ is tangent to the curve $\mathbf{r}(t)$ and that the vector

单位切向量:
$$\mathbf{T} = \frac{\mathbf{v}}{|\mathbf{v}|}$$

is therefore a unit vector tangent to the (smooth) curve, called the **unit tangent vector**



曲率与单位法向量

DEFINITION If $\mathbf{T}$ is the unit vector of a smooth curve, the **curvature** function of the curve is

曲率:
$$\kappa = \left| \frac{d\mathbf{T}}{ds} \right|. \quad \kappa = \frac{1}{|\mathbf{v}|} \left| \frac{d\mathbf{T}}{dt} \right|,$$

DEFINITION At a point where $\kappa \neq 0$, the **principal unit normal** vector for a smooth curve in the plane is

单位法向量:
$$\mathbf{N} = \frac{1}{\kappa} \frac{d\mathbf{T}}{ds}. \quad \mathbf{N} = \frac{d\mathbf{T}/dt}{|d\mathbf{T}/dt|},$$

曲率圆方程

求曲率圆方程: $y = f(x)$

$$\vec{r}(t) = t\hat{i} + f(t)\hat{j} \quad \vec{v}(t) = \hat{i} + f'(t)\hat{j}$$

$$\vec{T}(t) = \frac{\vec{v}}{|\vec{v}|} = \frac{\hat{i} + f'(t)\hat{j}}{\sqrt{1+f'(t)^2}} \quad \vec{T}'(t) = \frac{\sqrt{1+f'(t)^2}f''(t)\hat{j} - \frac{f'(t)f''(t)}{\sqrt{1+f'(t)^2}}(\hat{i} + f'(t)\hat{j})}{1+f'(t)^2}$$
$$= \frac{-f'(t)f''(t)\hat{i} + f''(t)\hat{j}}{(1+f'(t)^2)^{\frac{3}{2}}}$$

$$k(t) = \frac{1}{|\vec{v}(t)|}|\vec{T}'(t)| = \frac{f''(t)}{(1+f'(t)^2)^{\frac{3}{2}}} \quad R = \frac{1}{k(t)} = \frac{(1+f'(t)^2)^{\frac{3}{2}}}{f''(t)}$$

$$x_D = x_0 - R \cdot \frac{f'(t)}{\sqrt{1+f'(t)^2}} \quad y_D = y_0 + R \cdot \frac{1}{\sqrt{1+f'(t)^2}}$$

EXAMPLE 4 Find and graph the osculating circle of the parabola $y = x^2$ at the origin.

Chapter14

偏导数

DEFINITIONS Suppose D is a set of n -tuples of real numbers $(x_1, x_2, \dots, x_n)$. A **real-valued function** f on D is a rule that assigns a unique (single) real number

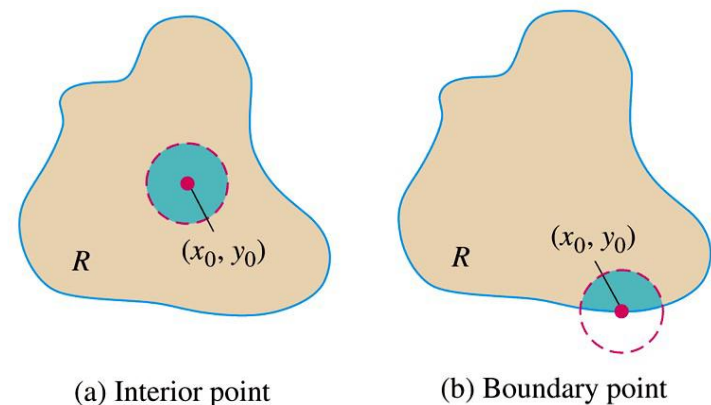
$$w = f(x_1, x_2, \dots, x_n)$$

to each element in D . The set D is the function's **domain**. The set of w -values taken on by f is the function's **range**. The symbol w is the **dependent variable** of f , and f is said to be a function of the n **independent variables** x_1 to x_n . We also call the x_j 's the function's **input variables** and call w the function's **output variable**.

二元函数的点

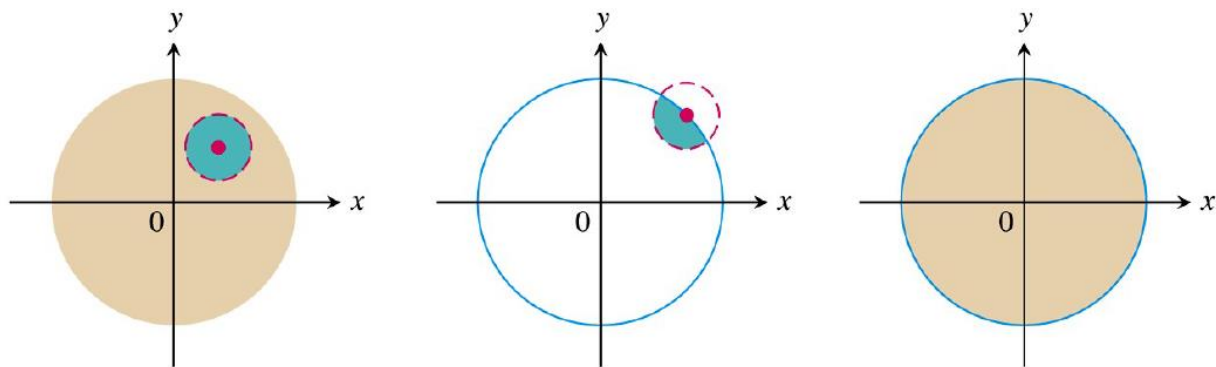
DEFINITIONS A point (x_0, y_0) in a region (set) R in the xy -plane is an **interior point** of R if it is the center of a disk of positive radius that lies entirely in R (Figure 14.2). A point (x_0, y_0) is a **boundary point** of R if every disk centered at (x_0, y_0) contains points that lie outside of R as well as points that lie in R . (The boundary point itself need not belong to R .)

The interior points of a region, as a set, make up the **interior** of the region. The region's boundary points make up its **boundary**. A region is **open** if it consists entirely of interior points. A region is **closed** if it contains all its boundary points (Figure 14.3).



内点； 边界点； 开放； 封闭

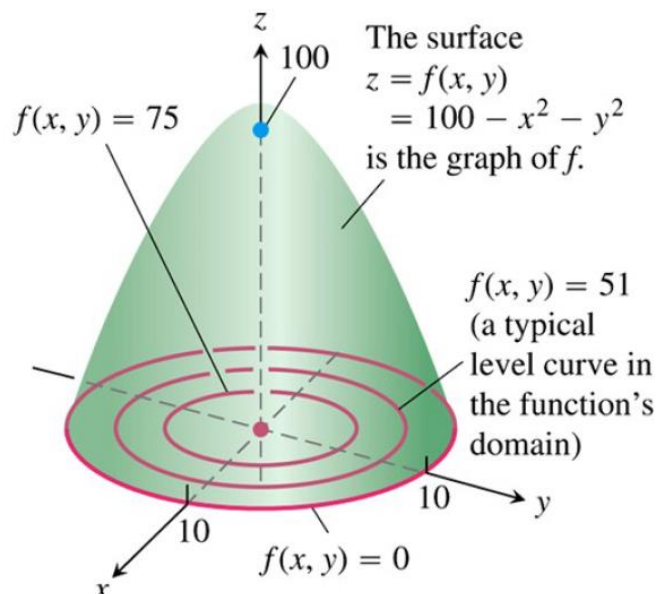
有界 VS. 无界



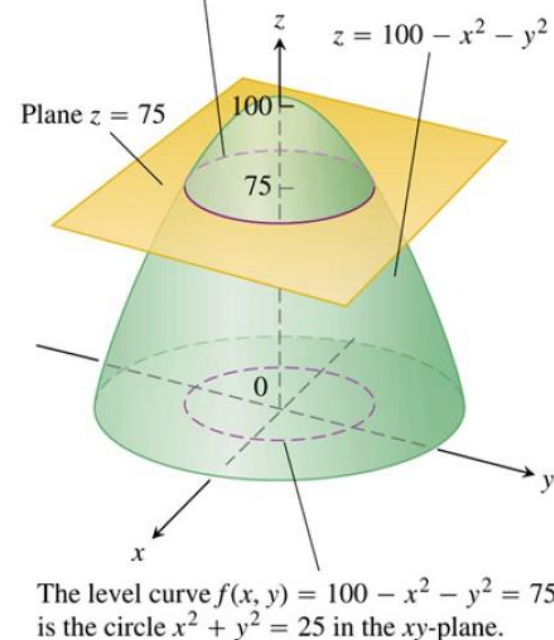
DEFINITIONS A region in the plane is **bounded** if it lies inside a disk of fixed radius. A region is **unbounded** if it is not bounded.

等高线与曲面

DEFINITIONS The set of points in the plane where a function $f(x, y)$ has a constant value $f(x, y) = c$ is called a **level curve** of f . The set of all points $(x, y, f(x, y))$ in space, for (x, y) in the domain of f , is called the **graph** of f . The graph of f is also called the **surface** $z = f(x, y)$.



The contour curve $f(x, y) = 100 - x^2 - y^2 = 75$ is the circle $x^2 + y^2 = 25$ in the plane $z = 75$.



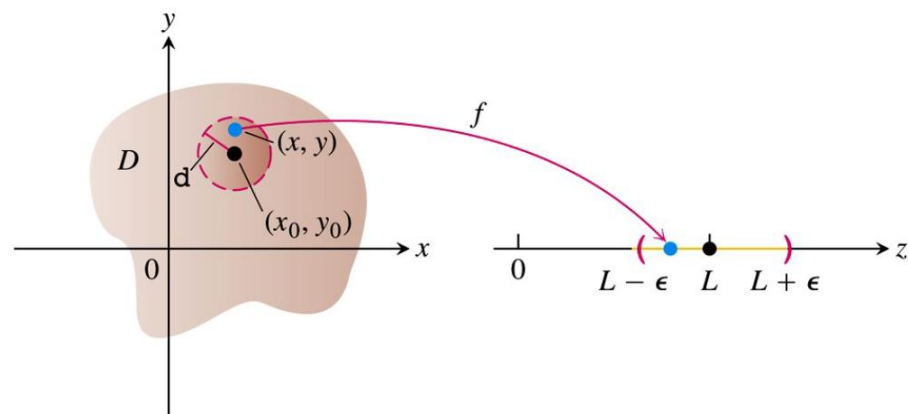
多元函数的极限

DEFINITION We say that a function $f(x, y)$ approaches the **limit** L as (x, y) approaches (x_0, y_0) , and write

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = L$$

if, for every number $\epsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all (x, y) in the domain of f ,

$$|f(x, y) - L| < \epsilon \quad \text{whenever} \quad 0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta.$$



多元函数极限运算法则

THEOREM 1—Properties of Limits of Functions of Two Variables

The following rules hold if L , M , and k are real numbers and

The fol-

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L \quad \text{and} \quad \lim_{(x,y) \rightarrow (x_0,y_0)} g(x,y) = M.$$

1. *Sum Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) + g(x,y)) = L + M$$

2. *Difference Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) - g(x,y)) = L - M$$

3. *Constant Multiple Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} kf(x,y) = kL \quad (\text{any number } k)$$

4. *Product Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) \cdot g(x,y)) = L \cdot M$$

5. *Quotient Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M}, \quad M \neq 0$$

6. *Power Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y)]^n = L^n, \quad n \text{ a positive integer}$$

7. *Root Rule:*

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \sqrt[n]{f(x,y)} = \sqrt[n]{L} = L^{1/n},$$

n a positive integer, and if n is even, we assume that $L > 0$.

Find $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}}.$

Find $\lim_{(x,y) \rightarrow (0,0)} \frac{4xy^2}{x^2 + y^2}$ if it exists.

求极限 $\lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{(y-x)x}{\sqrt{x^2 + y^2}}.$

多元函数的连续性

DEFINITION A function $f(x, y)$ is **continuous at the point** (x_0, y_0) if

1. f is defined at (x_0, y_0) ,
2. $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ exists,
3. $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = f(x_0, y_0)$.

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

A function is **continuous** if it is continuous at every point of its domain.

Two-Path Test for Nonexistence of a Limit

If a function $f(x, y)$ has different limits along two different paths in the domain of f as (x, y) approaches (x_0, y_0) , then $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ does not exist.

Having the same limit along all straight lines approaching (x_0, y_0) does not imply a limit exists at (x_0, y_0) .

$$f(x, y) = \frac{2x^2y}{x^4 + y^2}$$

排除法！

Continuity of Composites

If f is continuous at (x_0, y_0) and g is a single-variable function continuous at $f(x_0, y_0)$, then the composite function $h = g \circ f$ defined by $h(x, y) = g(f(x, y))$ is continuous at (x_0, y_0) .



SUSTech Southern University
of Science and
Technology

The background of the slide is composed of a network of interconnected nodes and lines, forming a complex, organic shape on the left and a more regular, triangular pattern on the right. The nodes are represented by small orange dots, and the lines are thin orange or grey lines. The overall color scheme is light grey and white, with orange accents.

Q&A

祝大家期中顺利！